INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Mathematics

Branch:- CSE/IT/ECE/MTC/PHY **Semester-I**

End-Term Examination – December 2025

Subject Name / Code: Eng. Math–I, CSE/IT/ECE/MTC/PHY–1001

Max. Marks: 60
Time Allotted: 03 Hrs

NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

S. No.	Question	Marks
1.	(a) Determine the value of α for which $\sum_{n=1}^{\infty} \frac{(-1)^n n^\alpha}{\sqrt{n+1}}$ is absolutely convergent, conditionally convergent, or divergent.	5
	(b) Evaluate $\iiint \sqrt{x^2+y^2}\,dxdydz$ over the volume bounded by the right circular cone $x^2+y^2=z^2, z>0$ and the planes $z=0, z=1$.	5
2.	(a) Evaluate $\iint e^{\frac{y-x}{y+x}}dxdy$ over the region $x=0, y=0$ and $x+y=2$.	5
	(b) Prove that $$\int_a^b (x-a)^m(b-x)^n dx = (b-a)^{m+n+1}\beta(m+1,n+1).$$	5
3.	(a) Prove that the function $f(x,y)=\sqrt{\mid xy \mid}$ is not differentiable at $(0,0)$, but f_x and f_y both exist at the origin and have the value 0.	5
	(b) The sum of the surfaces of a sphere and a cube is given. Show that when the sum of the volumes is least, the diameter of the sphere is equal to the edge of the cube.	5

S. No.	Question	Marks
4.	(a) Solve $\frac{dy}{dx} = \frac{y}{x+y\sin^2(\frac{x}{y})}$.	5
	(b) Solve $(xy^2 - x^2)dx + (3x^2y^2 + x^2y - 2x^3 + y^2)dy = 0$.	5
5.	(a) Solve $xy' - (\ln x)y^2 + y = 0, x > 0$	5
	(b) Solve $y'' - 3y' + 2y = \cos(e^{-x})$	5
6.	(a) Find the general solution of $$(2x+1)^2y'' + (2x+1)y' + y = 2x + 1.$$	5
	(b) Solve the given system of ordinary differential equations $$x' = 2x + y + e^t, \; y' = x + 3y + t.$$	5

ALL THE BEST